CYBERSECURITY INTERNSHIP

Organization name: Elevate labs

TASK - 2: Operating System Security Fundamentals (Linux & Windows)

Operating System Security Checklist

1. Operating System Used

- I used Kali Linux for this task.
- The system was running inside a Virtual Machine.

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2. User Accounts and Access Control

- I checked the current logged-in user using the terminal.
- I learned that Linux has different user types such as root (administrator) and normal users.
- I understood that administrator users have full system access, while normal users have limited permissions.
- I observed that admin access should be used only when required.

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3. File Permissions ‘chmod’

- I checked file permissions using the command ls -l.

- I learned about **read, write, and execute** permissions.
- I used the command chmod 700 filename to give full permissions to the file owner and remove access from others.
- I understood that this helps protect files from unauthorized access.

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4. Administrator vs Standard User

- I understood the difference between administrator and standard users.
- I observed that system-level changes require administrator (sudo) access.
- This helps prevent accidental or malicious changes to the system.

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5. Firewall Configuration

- I learned that UFW is not installed by default in Kali Linux.
- I checked firewall rules using the command iptables -L.
- I understood that firewall rules control incoming and outgoing network traffic.

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6. Running Processes and Services

- I viewed running processes using commands like ps and top.
- I learned that many services run in the background.
- I understood that unnecessary services can increase security risks.

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7. Reducing Attack Surface

- I learned that disabling unused services reduces the attack surface.
- Fewer running services mean fewer chances for attackers to exploit the system.

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8. OS Hardening Practices Learned

- Use strong passwords
- Limit administrator access
- Manage file permissions properly
- Monitor running processes
- Enable or manage firewall rules

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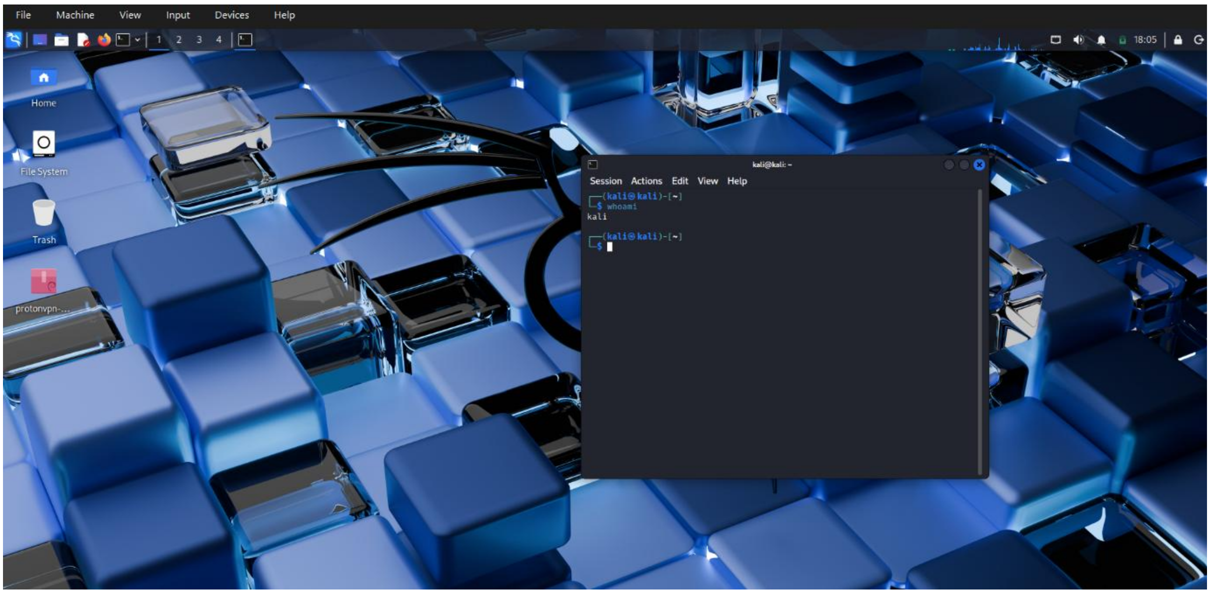
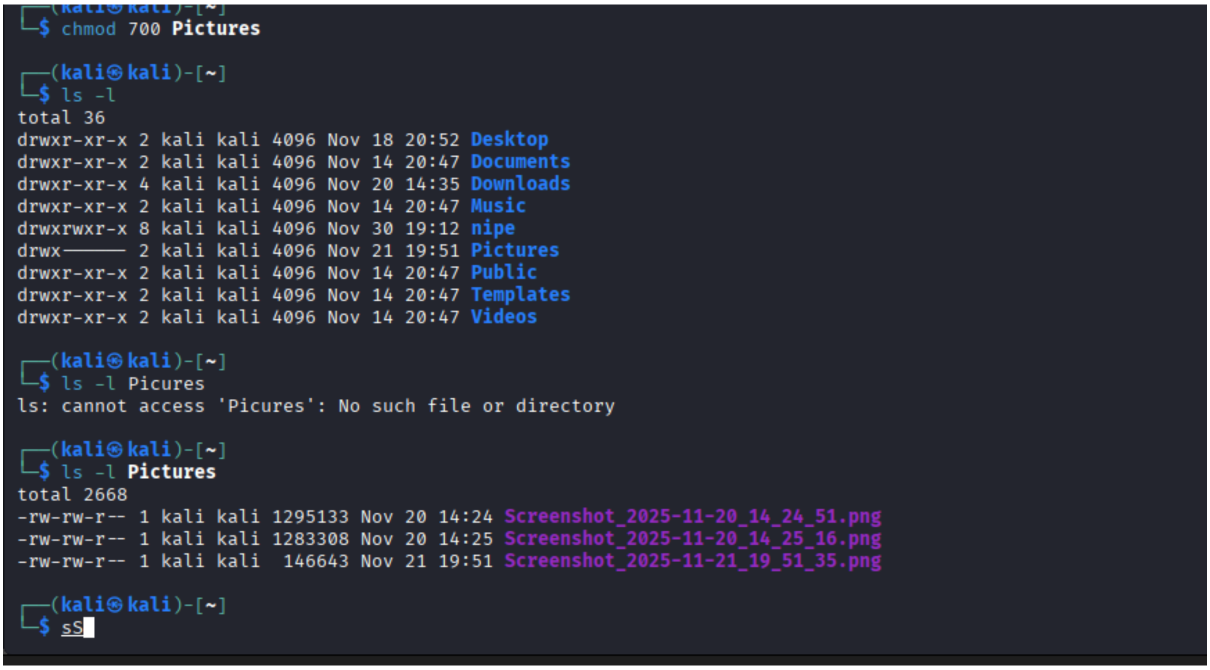
9. Conclusion

Through this task, I gained a clear understanding of operating system security at different levels such as users, files, and networks. I learned how user permissions, firewalls, and system services play an important role in keeping the system secure. This task helped me understand the basics of OS hardening and strengthened my knowledge of system security.

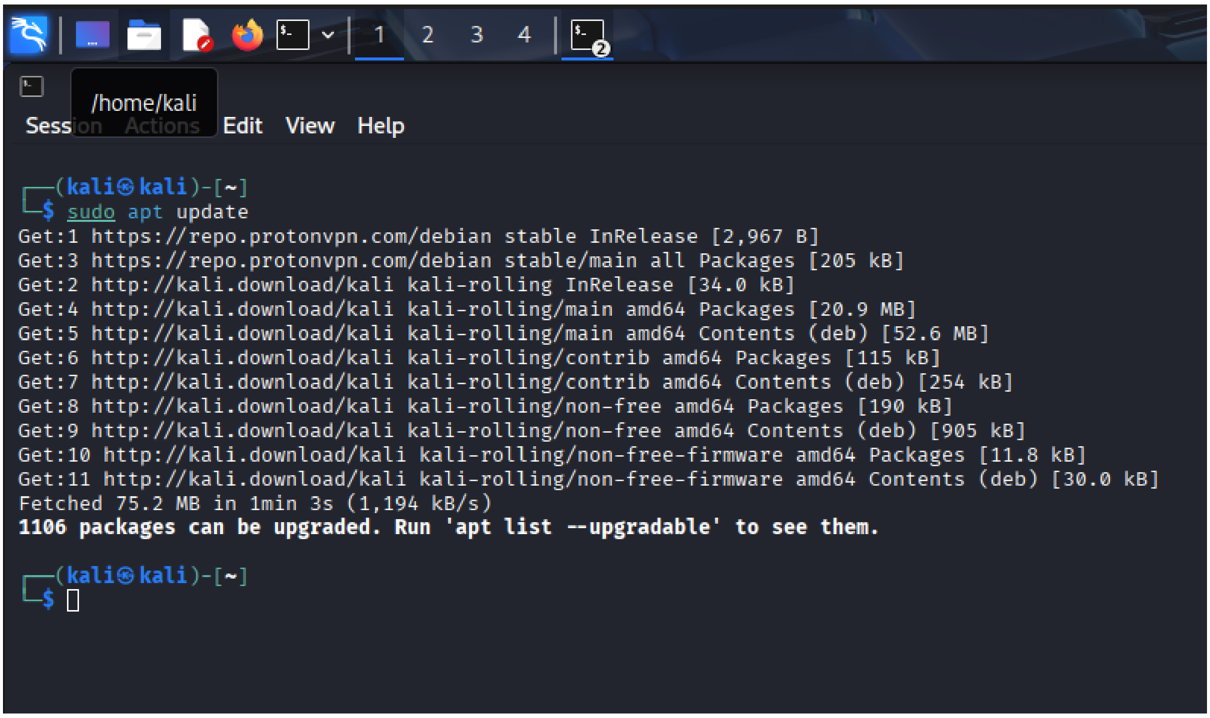

File Machine View Input Devices Help
18:05
Home
File System
Trash
protonvpn-...
kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ whoami
kali
┌──(kali㉿kali)-[~]
└─$


File Machine View Input Devices Help
18:06
kali@kali: ~
Session Actions Edit View Help
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
mysql:x:101:102:MariaDB Server:/nonexistent:/bin/false
tss:x:102:104:TPM software stack:/var/lib/tpm:/bin/false
strongswan:x:103:65534::/var/lib/strongswan:/usr/sbin/nologin
systemd-timesync:x:991:991:systemd Time Synchronization:/:/usr/sbin/nologin
_gophish:x:104:106::/var/lib/gophish:/usr/sbin/nologin
iodine:x:105:65534::/run/iodine:/usr/sbin/nologin
messagebus:x:990:990:System Message Bus:/nonexistent:/usr/sbin/nologin
tcpdump:x:106:107::/nonexistent:/usr/sbin/nologin
miredo:x:107:65534::/var/run/miredo:/usr/sbin/nologin
_rpc:x:108:65534::/run/rpcbind:/usr/sbin/nologin
redis:x:109:110::/var/lib/redis:/usr/sbin/nologin
mosquitto:x:110:113::/var/lib/mosquitto:/usr/sbin/nologin
redsocks:x:111:114::/var/run/redsocks:/usr/sbin/nologin
stunnel4:x:989:989:stunnel service system account:/var/run/stunnel4:/usr/sbin/nologin
sshd:x:988:65534:sshd user:/run/sshd:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
Debian-snmp:x:112:115::/var/lib/snmp:/bin/false
sslh:x:113:117::/nonexistent:/usr/sbin/nologin
postgres:x:114:118:PostgreSQL administrator:/var/lib/postgresql:/bin/bash
avahi:x:115:119:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin
speech-dispatcher:x:116:29:Speech Dispatcher:/run/speech-dispatcher:/bin/false
_gvm:x:117:120::/var/lib/openvas:/usr/sbin/nologin
usbmux:x:118:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
cups-pk-helper:x:119:121:user for cups-pk-helper service:/nonexistent:/usr/sbin/nologin
nm-openvpn:x:120:122:NetworkManager OpenVPN:/var/lib/openvpn/chroot:/usr/sbin/nologin
inetsim:x:121:123::/var/lib/inetsim:/usr/sbin/nologin
pipewire:x:986:986:system user for pipewire:/nonexistent:/usr/sbin/nologin
nm-openconnect:x:122:125:NetworkManager OpenConnect plugin:/var/lib/NetworkManager:/usr/sbin/nologin
geoclue:x:123:126::/var/lib/geoclue:/usr/sbin/nologin
lightdm:x:124:127:Light Display Manager:/var/lib/lightdm:/bin/false
statd:x:125:65534::/var/lib/nfs:/usr/sbin/nologin
saned:x:126:128::/var/lib/saned:/usr/sbin/nologin
polkitd:x:985:985:User for polkitd:/:/usr/sbin/nologin
rtkit:x:127:129:RealtimeKit:/proc:/usr/sbin/nologin
colord:x:128:130:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
kali:x:1000:1000:Kali,,,:/home/kali:/usr/bin/zsh
pcscd:x:984:984:PC/SC Smart Card Daemon:/:/usr/sbin/nologin
debian-tor:x:129:134::/var/lib/tor:/bin/false
┌──(kali㉿kali)-[~]
└─$ sudo cpan m --installdeps

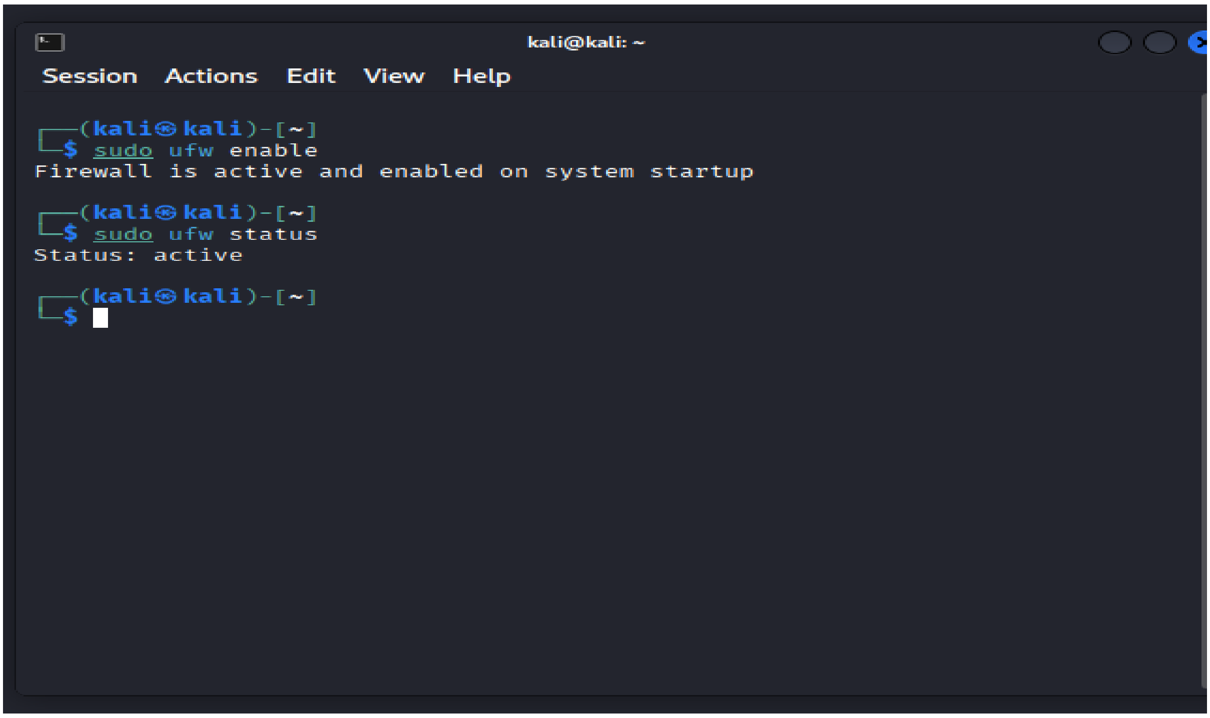

rtkit:x:127:129:RealtimeKit:/proc:/usr/sbin/nologin
colord:x:128:130:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
kali:x:1000:1000:Kali,,,:/home/kali:/usr/bin/zsh
pcscd:x:984:984:PC/SC Smart Card Daemon:/:/usr/sbin/nologin
debian-tor:x:129:134::/var/lib/tor:/bin/false
┌──(kali㉿kali)-[~]
└─$ ls -l
total 36
drwxr-xr-x 2 kali kali 4096 Nov 18 20:52 Desktop
drwxr-xr-x 2 kali kali 4096 Nov 14 20:47 Documents
drwxr-xr-x 4 kali kali 4096 Nov 20 14:35 Downloads
drwxr-xr-x 2 kali kali 4096 Nov 14 20:47 Music
drwxrwxr-x 8 kali kali 4096 Nov 30 19:12 nipe
drwxr-xr-x 2 kali kali 4096 Nov 21 19:51 Pictures
drwxr-xr-x 2 kali kali 4096 Nov 14 20:47 Public
drwxr-xr-x 2 kali kali 4096 Nov 14 20:47 Templates
drwxr-xr-x 2 kali kali 4096 Nov 14 20:47 Videos
┌──(kali㉿kali)-[~]
└─$ ls -l
total 36
drwxr-xr-x 2 kali kali 4096 Nov 18 20:52 Desktop
drwxr-xr-x 2 kali kali 4096 Nov 14 20:47 Documents
drwxr-xr-x 4 kali kali 4096 Nov 20 14:35 Downloads
drwxr-xr-x 2 kali kali 4096 Nov 14 20:47 Music
drwxrwxr-x 8 kali kali 4096 Nov 30 19:12 nipe
drwxr-xr-x 2 kali kali 4096 Nov 21 19:51 Pictures
drwxr-xr-x 2 kali kali 4096 Nov 14 20:47 Public
drwxr-xr-x 2 kali kali 4096 Nov 14 20:47 Templates
drwxr-xr-x 2 kali kali 4096 Nov 14 20:47 Videos
┌──(kali㉿kali)-[~]
└─$ sS

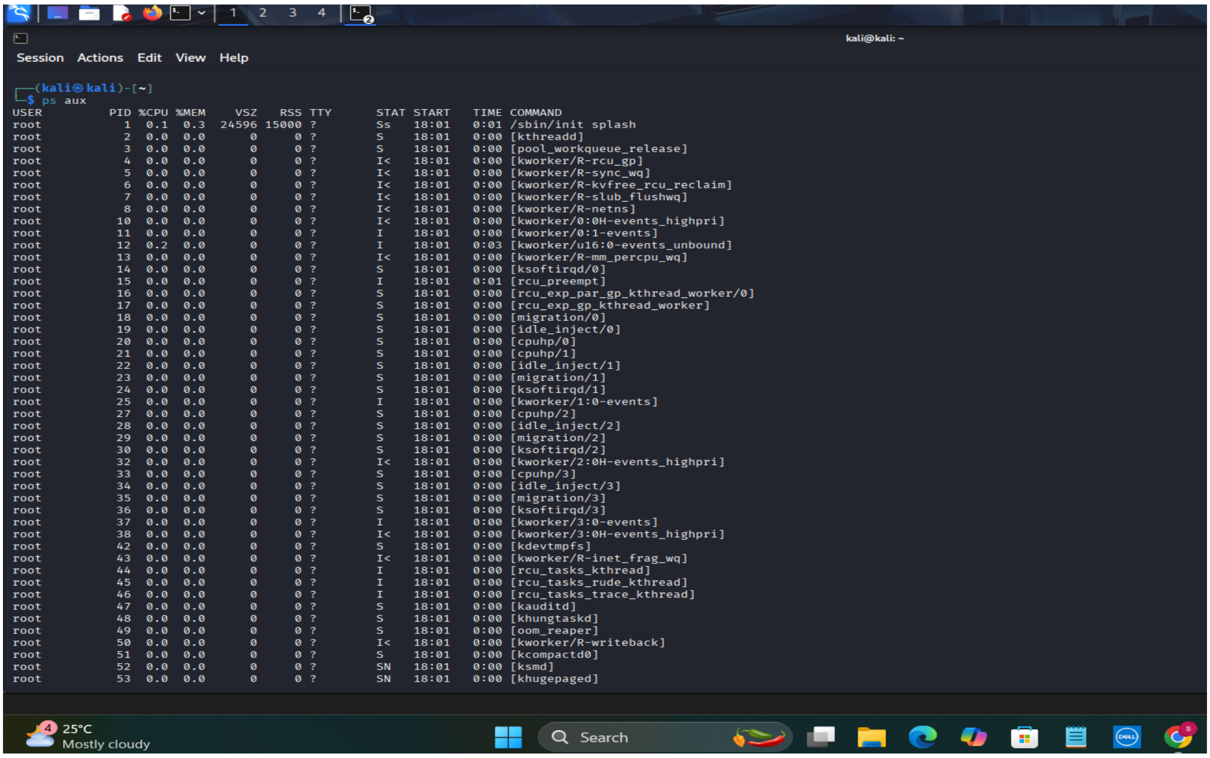


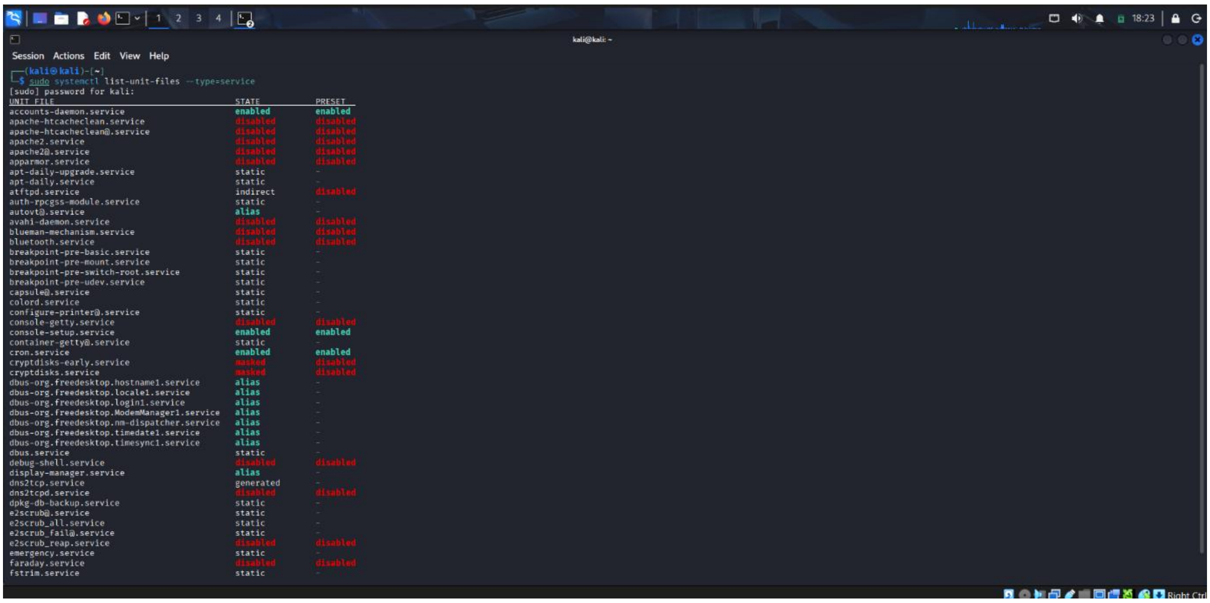
```
└─$ chmod 700 Pictures

┌──(kali㉿kali)-[~]
└─$ ls -l
total 36
drwxr-xr-x 2 kali kali 4096 Nov 18 20:52 Desktop
drwxr-xr-x 2 kali kali 4096 Nov 14 20:47 Documents
drwxr-xr-x 4 kali kali 4096 Nov 20 14:35 Downloads
drwxr-xr-x 2 kali kali 4096 Nov 14 20:47 Music
drwxrwxr-x 8 kali kali 4096 Nov 30 19:12 nipe
drwx------ 2 kali kali 4096 Nov 21 19:51 Pictures
drwxr-xr-x 2 kali kali 4096 Nov 14 20:47 Public
drwxr-xr-x 2 kali kali 4096 Nov 14 20:47 Templates
drwxr-xr-x 2 kali kali 4096 Nov 14 20:47 Videos

┌──(kali㉿kali)-[~]
└─$ ls -l Picures
ls: cannot access 'Picures': No such file or directory

┌──(kali㉿kali)-[~]
└─$ ls -l Pictures
total 2668
-rw-rw-r-- 1 kali kali 1295133 Nov 20 14:24 Screenshot_2025-11-20_14_24_51.png
-rw-rw-r-- 1 kali kali 1283308 Nov 20 14:25 Screenshot_2025-11-20_14_25_16.png
-rw-rw-r-- 1 kali kali  146643 Nov 21 19:51 Screenshot_2025-11-21_19_51_35.png

┌──(kali㉿kali)-[~]
└─$ sS
```




```
┌──(kali㉿kali)-[~]
└─$ sudo apt update
Get:1 https://repo.protonvpn.com/debian stable InRelease [2,967 B]
Get:3 https://repo.protonvpn.com/debian stable/main all Packages [205 kB]
Get:2 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 Packages [20.9 MB]
Get:5 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.6 MB]
Get:6 http://kali.download/kali kali-rolling/contrib amd64 Packages [115 kB]
Get:7 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [254 kB]
Get:8 http://kali.download/kali kali-rolling/non-free amd64 Packages [190 kB]
Get:9 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [905 kB]
Get:10 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [11.8 kB]
Get:11 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [30.0 kB]
Fetched 75.2 MB in 1min 3s (1,194 kB/s)
1106 packages can be upgraded. Run 'apt list --upgradable' to see them.

┌──(kali㉿kali)-[~]
└─$ 
```


kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ sudo ufw enable
Firewall is active and enabled on system startup
(kali㉿kali)-[~]
$ sudo ufw status
Status: active
(kali㉿kali)-[~]
$


1 2 3 4
kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ ps aux
USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND
root 1 0.1 0.3 24596 15000 ? Ss 18:01 0:01 /sbin/init splash
root 2 0.0 0.0 0 0 ? S 18:01 0:00 [kthreadd]
root 3 0.0 0.0 0 0 ? S 18:01 0:00 [pool_workqueue_release]
root 4 0.0 0.0 0 0 ? I< 18:01 0:00 [kworker/R-rcu_gp]
root 5 0.0 0.0 0 0 ? I< 18:01 0:00 [kworker/R-sync_wq]
root 6 0.0 0.0 0 0 ? I< 18:01 0:00 [kworker/R-kvfree_rcu_reclaim]
root 7 0.0 0.0 0 0 ? I< 18:01 0:00 [kworker/R-slub_flushwq]
root 8 0.0 0.0 0 0 ? I< 18:01 0:00 [kworker/R-netns]
root 10 0.0 0.0 0 0 ? I< 18:01 0:00 [kworker/0:0H-events_highpri]
root 11 0.0 0.0 0 0 ? I 18:01 0:00 [kworker/0:1-events]
root 12 0.2 0.0 0 0 ? I 18:01 0:03 [kworker/u16:0-events_unbound]
root 13 0.0 0.0 0 0 ? I< 18:01 0:00 [kworker/R-mm_percpu_wq]
root 14 0.0 0.0 0 0 ? S 18:01 0:00 [ksoftirqd/0]
root 15 0.0 0.0 0 0 ? I 18:01 0:01 [rcu_preempt]
root 16 0.0 0.0 0 0 ? S 18:01 0:00 [rcu_exp_par_gp_kthread_worker/0]
root 17 0.0 0.0 0 0 ? S 18:01 0:00 [rcu_exp_gp_kthread_worker]
root 18 0.0 0.0 0 0 ? S 18:01 0:00 [migration/0]
root 19 0.0 0.0 0 0 ? S 18:01 0:00 [idle_inject/0]
root 20 0.0 0.0 0 0 ? S 18:01 0:00 [cpuhp/0]
root 21 0.0 0.0 0 0 ? S 18:01 0:00 [cpuhp/1]
root 22 0.0 0.0 0 0 ? S 18:01 0:00 [idle_inject/1]
root 23 0.0 0.0 0 0 ? S 18:01 0:00 [migration/1]
root 24 0.0 0.0 0 0 ? S 18:01 0:00 [ksoftirqd/1]
root 25 0.0 0.0 0 0 ? I 18:01 0:00 [kworker/1:0-events]
root 27 0.0 0.0 0 0 ? S 18:01 0:00 [cpuhp/2]
root 28 0.0 0.0 0 0 ? S 18:01 0:00 [idle_inject/2]
root 29 0.0 0.0 0 0 ? S 18:01 0:00 [migration/2]
root 30 0.0 0.0 0 0 ? S 18:01 0:00 [ksoftirqd/2]
root 32 0.0 0.0 0 0 ? I< 18:01 0:00 [kworker/2:0H-events_highpri]
root 33 0.0 0.0 0 0 ? S 18:01 0:00 [cpuhp/3]
root 34 0.0 0.0 0 0 ? S 18:01 0:00 [idle_inject/3]
root 35 0.0 0.0 0 0 ? S 18:01 0:00 [migration/3]
root 36 0.0 0.0 0 0 ? S 18:01 0:00 [ksoftirqd/3]
root 37 0.0 0.0 0 0 ? I 18:01 0:00 [kworker/3:0-events]
root 38 0.0 0.0 0 0 ? I< 18:01 0:00 [kworker/3:0H-events_highpri]
root 42 0.0 0.0 0 0 ? S 18:01 0:00 [kdevtmpfs]
root 43 0.0 0.0 0 0 ? I< 18:01 0:00 [kworker/R-inet_frag_wq]
root 44 0.0 0.0 0 0 ? I 18:01 0:00 [rcu_tasks_kthread]
root 45 0.0 0.0 0 0 ? I 18:01 0:00 [rcu_tasks_rude_kthread]
root 46 0.0 0.0 0 0 ? I 18:01 0:00 [rcu_tasks_trace_kthread]
root 47 0.0 0.0 0 0 ? S 18:01 0:00 [kauditd]
root 48 0.0 0.0 0 0 ? S 18:01 0:00 [khungtaskd]
root 49 0.0 0.0 0 0 ? S 18:01 0:00 [oom_reaper]
root 50 0.0 0.0 0 0 ? I< 18:01 0:00 [kworker/R-writeback]
root 51 0.0 0.0 0 0 ? S 18:01 0:00 [kcompactd0]
root 52 0.0 0.0 0 0 ? SN 18:01 0:00 [ksmd]
root 53 0.0 0.0 0 0 ? SN 18:01 0:00 [khugepaged]
25°C
Mostly cloudy
Search



kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ sudo systemctl list-unit-files --type=service
[sudo] password for kali:
UNIT FILE STATE PRESET
accounts-daemon.service enabled enabled
apache-htcacheclean.service disabled disabled
apache-htcacheclean@.service disabled disabled
apache2.service disabled disabled
apache2@.service disabled disabled
apparmor.service disabled disabled
apt-daily-upgrade.service static -
apt-daily.service static -
atftpd.service indirect disabled
auth-rpcgss-module.service static -
autovt@.service alias -
avahi-daemon.service disabled disabled
blueman-mechanism.service disabled disabled
bluetooth.service disabled disabled
breakpoint-pre-basic.service static -
breakpoint-pre-mount.service static -
breakpoint-pre-switch-root.service static -
breakpoint-pre-udev.service static -
capsule@.service static -
colord.service static -
configure-printer@.service static -
console-getty.service disabled disabled
console-setup.service enabled enabled
container-getty@.service static -
cron.service enabled enabled
cryptdisks-early.service masked disabled
cryptdisks.service masked disabled
dbus-org.freedesktop.hostname1.service alias -
dbus-org.freedesktop.locale1.service alias -
dbus-org.freedesktop.login1.service alias -
dbus-org.freedesktop.ModemManager1.service alias -
dbus-org.freedesktop.nm-dispatcher.service alias -
dbus-org.freedesktop.timedate1.service alias -
dbus-org.freedesktop.timesync1.service alias -
dbus.service static -
debug-shell.service disabled disabled
display-manager.service alias -
dns2tcp.service generated -
dns2tcpd.service disabled disabled
dpkg-db-backup.service static -
e2scrub@.service static -
e2scrub_all.service static -
e2scrub_fail@.service static -
e2scrub_reap.service disabled disabled
emergency.service static -
faraday.service disabled disabled
fstrim.service static -
18:23
Right Ctrl